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Data Structures and Algorithms II
Project 3
User's Manual

Setup and Compilation

1. Download and unzip the submission from eLearning on a Linux box in the multi-platform lab.
2. The submission includes:
    - AdjacencyMatrix.hpp
    - AdjacencyMatrix.cpp
    - BruteForce.hpp
    - BruteForce.cpp
    - GeneticAlgorithm.hpp
    - GeneticAlgorithm.cpp
    - TSP.hpp
    - TSP.cpp
    - main.cpp
    - Makefile
    - proj3 (executable)
    - UMLDiagram.png
    - UsersManual.pdf (this file)
    - distances.txt
    - Results.xlsx
3. Environment: This program has been tested on the UWF SSH Server and will run there.
4. Compiling. This program includes a `Makefile`. At the command line, navigate into the `src` directory. Once inside the src directory, type `make`. The program produces an executable entitled `proj3`.

Running the program: Be sure that `distances.txt` is located in the project's root directory (above the `src` directory; this file is in the same folder). Navigate back into the `src` directory. Issue the command `./proj3`. No command-line arguments are required or checked.

User input: A user will be prompted with four questions at the beginning of the program.

1) "How many cities to run?"
    a) Enter an integer value greater than 0 and less than or equal to 20.
2) "How many individual tours are there in a given generation?"
    a) Enter an integer value greater than 0.
3) "How many generations to run?"
    a) Enter an integer value greater than 0.
4) "What percentage of a generation should be comprised of mutations? Write in Decimal Format (i.e. If you want a mutation rate of 25%, input 0.25)"
    a) Enter a floating point value greater than 0 and less than or equal to 1.

Output: All output goes to the console. Output will be similar to this:

Brute Force Most Optimal Tour: 0 1 5 7 11 2 6 8 4 10 3 9 0 | Distance (cost): 424.88 | Run-time for Brute Force: 5.95986
Genetic Algorithm Most Optimal Tour: 0 9 11 2 6 7 3 8 4 10 1 5 0 | Distance (cost): 577.42 | Run-time for Genetic Algorithm: 0.0281799